From GD to fSGLD: Designing Noise in Stochastic Optimization

A Reading Report

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Abstract

This report consolidates my reading of two documents: Royer's lecture notes on stochastic gradient methods [1], and the paper by Bruno et al. introducing Flatness-Aware Stochastic Gradient Langevin Dynamics (fSGLD) [2]. It reviews the motivation, working principle, and strengths and weaknesses of GD, SGD, SGLD, and fSGLD, with the emphasis on fSGLD: by simply evaluating the gradient at a randomly perturbed point, it acquires an implicit Hessian-trace regularization and hence prefers flat minima at essentially no additional cost. My overall impression is that the four algorithms trace an evolution in the attitude toward noise: no noise, noise tolerated, noise designed, and finally noise endowed with geometric structure.

Keywords: stochastic gradient descent; Langevin dynamics; Gibbs distribution; flat minima; Hessian-trace regularization

1 Introduction and Notation

The central computational problem of deep learning is empirical risk minimization. The two documents handle the same skeleton—*parameter = parameter − stepsize × gradient estimate (+ noise)*—but from different standpoints: the lecture notes [1] develop the classical convergence theory of GD and SGD, while the fSGLD paper [2] asks how to sample solutions that are simultaneously deep and flat on a non-convex loss surface. The report follows the order GD → SGD → SGLD → fSGLD; each stage is the foundation of the next.

Notation (uniform throughout). To keep the notation consistent, I adopt that of the fSGLD paper and translate the lecture notes into it. Parameters are written $\theta \in \mathbb{R}^d$ (the notes use x). The objective is

$$u(\theta) = \mathbb{E}\big[U(\theta, X)\big], \tag{1}$$

where $U(\theta, X)$ is its stochastic version and X the data random variable (the notes use f and f_i; the finite-sum case $u = \frac{1}{n}\sum_{i=1}^{n} u_i$ corresponds to X uniform on $\{1, \dots, n\}$). The stepsize is λ throughout (the notes' α and ω_k). The variance proxy of the

gradient estimator is ρ^2, i.e. $\mathbb{E}\|U(\theta, X)\|^2 \leq \rho^2 + \|\nabla u(\theta)\|^2$ (the notes write φ^2; I rename it ρ^2 to reserve φ for the perturbation of fSGLD). Also used below: L for the smoothness constant, μ for strong convexity, $H(\theta)$ for the Hessian, β for the inverse temperature, and σ for the perturbation scale.

2 GD and SGD: Classical Stochastic Optimization

2.1 GD: The Deterministic Baseline

GD iterates along the negative gradient, $\theta_{k+1} = \theta_k - \lambda \nabla u(\theta_k)$, scanning the full dataset at every step at a cost of $O(n)$ gradient evaluations. Its theory rests on the descent lemma supplied by L-smoothness:

$$u(\theta_{k+1}) \leq u(\theta_k) - \Big(\lambda - \frac{L\lambda^2}{2}\Big)\|\nabla u(\theta_k)\|^2, \tag{2}$$

so that for $\lambda \leq 1/L$ the objective decreases strictly at every step. The classical consequences [1] follow: geometric convergence to the exact solution in the strongly convex case, $O(1/K)$ in the convex case, and in the non-convex case only $\min_{k<K} \|\nabla u(\theta_k)\|^2 = O(1/K)$, i.e. convergence to some stationary point. GD is exact and deterministic, but its per-step cost is infeasible on large datasets, and on non-convex problems it has no mechanism for escaping bad local minima: which basin it ends up in is decided entirely by the initialization.

2.2 SGD: Randomization and Its Price

SGD replaces the full gradient by a single-sample (or mini-batch) estimator $U(\theta_k, X_{k+1})$ and, by unbiasedness, cuts the per-step cost by a factor of n. It is not a descent method, and its guarantees hold only in expectation:

$$\mathbb{E}\big[u(\theta_{k+1})\big] \leq u(\theta_k) - \Big(\lambda - \frac{L\lambda^2}{2}\Big)\|\nabla u(\theta_k)\|^2 + \frac{L\lambda^2}{2}\,\rho^2. \tag{3}$$

Compared with (2), the extra tail term is a *variance tax*: it never vanishes as the iterations proceed and is the root of all theoretical limitations of SGD. The four theorems of the notes (§2.3) can be summarized as follows. With a constant stepsize, the expected suboptimality on strongly convex problems converges only to a neighborhood of size about $\lambda L\rho^2/(2\mu)$—the *noise floor*; with a diminishing stepsize one can reach arbitrary accuracy, but the rate degrades to $O(1/K)$, which is already information-theoretically optimal for a stochastic first-order oracle. In other words, the variance tax reflects the difficulty of the problem rather than a defect of the algorithm; mini-batching, SVRG, and Adam are all ways of negotiating with it.

The noise floor admits another reading: with a constant stepsize, SGD oscillates near a minimizer and settles into a stationary distribution, the gradient noise acting as an *effective temperature*. This is precisely the seed of SGLD.

3 SGLD: Turning Noise into Temperature

SGLD [3, 4] adopts the opposite attitude: rather than eliminating noise, one injects deliberately designed noise. The motivation is twofold. From the Bayesian viewpoint one wants to sample the Gibbs distribution and average predictions, obtaining uncertainty estimates for free; from the non-convex optimization viewpoint, as $\beta \to \infty$ the Gibbs distribution concentrates on the global minimizers, so noise provides a theoretically grounded escape mechanism [4]. Operationally, the dynamics repeatedly escapes, falls back, and escapes again, distributing its time among the basins in the correct statistical proportion, so that the deepest and widest basin receives the longest dwell time. Combined with the noise-floor perspective, SGLD simply turns the implicit effective temperature of SGD into an explicit, tunable parameter β.

The starting point is the overdamped Langevin equation

$$dY_t = -\nabla u(Y_t)\, dt + \sqrt{2\beta^{-1}}\, dB_t, \tag{4}$$

whose invariant measure is exactly $\pi_\beta^{\mathrm{SGLD}} \propto e^{-\beta u}$. By the Fokker–Planck equation the density p_t satisfies $\partial_t p = \nabla\cdot(p\nabla u) + \beta^{-1}\Delta p$; setting the probability flux $p\nabla u + \beta^{-1}\nabla p$ to zero yields $\nabla \log p = -\beta\nabla u$, i.e. $p \propto e^{-\beta u}$. Conversely speaking, the noise scale $\sqrt{2\beta^{-1}}$ is chosen precisely so that the stationary balance of drift and diffusion lands on this Gibbs distribution. Applying an Euler discretization to (4), using $B_{t+\lambda} - B_t \sim \mathcal{N}(0, \lambda)$, and replacing the exact gradient by a stochastic one gives the SGLD iteration

$$\theta_{k+1}^{\mathrm{SGLD}} = \theta_k^{\mathrm{SGLD}} - \lambda\, U(\theta_k^{\mathrm{SGLD}}, X_{k+1}) + \sqrt{2\lambda\beta^{-1}}\, \xi_{k+1}, \qquad \xi_{k+1} \sim \mathcal{N}(0, I_d). \tag{5}$$

The strengths of SGLD are global exploration and a dual output: it serves as an optimizer and a posterior sampler at the same cost as SGD. Its weaknesses are slow mixing and a discretization bias under a constant stepsize. But the most decisive point—and the starting point of the fSGLD paper—is that its invariant measure $e^{-\beta u}$ is determined solely by objective values and has *no preference* for the geometry of the loss surface: a sharp basin and a flat basin of equal depth receive the same probability mass.

4 fSGLD: Noise with a Geometric Bias (Focus)

4.1 Motivation and Algorithm

The flat-minima hypothesis holds that solutions in low-curvature regions generalize better [5, 6], and a family of methods has grown from it—SAM [8], Entropy-SGD [7], Entropy-MCMC [9], and others. Each, however, pays a price: two gradient evaluations per step, auxiliary variables that double the memory, or the like; and all are essentially local—having landed in a basin, they only look for flatness within it. SGLD, on the other hand, explores globally but is blind to geometry: any preference it has for wide basins is only a passive volume (entropy) bonus in the invariant measure. fSGLD aims at both properties at no additional cost, turning that passive bonus into an explicit penalty written into the energy. It writes down the desired solutions as the Hessian-trace regularized objective

$$v(\theta) = u(\theta) + \frac{\sigma^2}{2} \operatorname{tr} H(\theta), \tag{6}$$

whose minimizers are called the global flat minima. Optimizing v directly requires second-order information, which is infeasible in high dimension, so fSGLD realizes it implicitly: the only difference from SGLD is that the gradient is evaluated at a perturbed point $\theta_k + \varphi_{k+1}$,

$$\theta_{k+1}^{\mathrm{fSGLD}} = \theta_k^{\mathrm{fSGLD}} - \lambda\, U\big(\theta_k^{\mathrm{fSGLD}} + \varphi_{k+1}, X_{k+1}\big) + \sqrt{2\lambda\beta^{-1}}\, \xi_{k+1}, \tag{7}$$

where $\varphi_{k+1} \sim \mathcal{N}(0, \sigma^2 I_d)$ is the new perturbation noise, independent of the Langevin noise ξ and of the data X. The perturbation scale is not a free hyperparameter but is fixed by the *coupling condition*

$$\sigma = \beta^{-(1+\eta)/4}, \qquad \eta \in (0, 1) \text{ fixed (the paper uses } 0.1), \tag{8}$$

so that fSGLD requires exactly as many hyperparameters as standard SGLD: only β. Unlike the mostly local guarantees of SAM-type methods, the paper's guarantees are global and non-asymptotic: they specify, at least in principle, how small the stepsize must be, how many iterations suffice, and how large the remaining error floor is.

4.2 Mechanism: Why a Perturbed Gradient Biases Toward Flatness

The argument in §2.3 of the paper proceeds in two steps. First, define the randomized-smoothing surrogate

$$g_\sigma(\theta) = \mathbb{E}_\varphi\big[u(\theta + \varphi)\big], \qquad \nabla g_\sigma(\theta) = \mathbb{E}_\varphi\big[\nabla u(\theta + \varphi)\big], \tag{9}$$

so a gradient evaluated at a perturbed point is an unbiased estimator of ∇g_σ, and fSGLD is simply standard SGLD run on g_σ. A useful mental image: standard SGLD senses the landscape with a single geometric point, whereas fSGLD rolls a ball of radius σ over it, and what the ball feels is the average loss over its footprint. Intuitively, g_σ is a blurred copy of u: a narrow, sharp basin is filled in by its high surroundings, while a wide basin survives almost unchanged, so optimizing g_σ automatically avoids narrow minima. Note that the perturbed gradient is biased with respect to the original objective u—and this bias is exactly the point.

Second, expand $u(\theta+\varphi)$ by Taylor's theorem and take the expectation over φ:

$$g_\sigma(\theta) = \underbrace{u(\theta)}_{\text{original objective}} + \underbrace{\frac{\sigma^2}{2}\operatorname{tr} H(\theta)}_{\text{curvature penalty}} + \mathbb{E}\big[R(\theta,\varphi)\big], \qquad \mathbb{E}[R] = O(\sigma^4 d^2). \tag{10}$$

The first-order term vanishes because $\mathbb{E}[\varphi]=0$; the third-order term vanishes because odd moments of a centered symmetric distribution are zero; and the second-order term is computed directly from independence: $\mathbb{E}[\varphi^\top H\varphi] = \sum_{i,j} H_{ij}\mathbb{E}[\varphi_i\varphi_j] = \sigma^2\sum_i H_{ii} = \sigma^2 \operatorname{tr} H$, which is precisely the regularizer in (6). Hence, as long as the remainder $\mathbb{E}[R]$ is small, fSGLD optimizes v implicitly without ever touching second-order information.

4.3 The Logic of the Coupling

In high dimension $\sigma^4 d^2$ is not a priori small: if $\mathbb{E}[R]$ is large, the blur no longer yields a clean curvature preference but a polluted mixture, so σ cannot be chosen freely. The logic of the coupling (8) is visible in two lines of arithmetic (the rigorous argument is in Appendix B of the paper): the invariant measure of fSGLD is $\pi_\beta^{\mathrm{fSGLD}} \propto e^{-\beta g_\sigma}$, the ideal target is $\pi_{\beta,\sigma}^{\phi} \propto e^{-\beta v}$, and their ratio is controlled by $e^{\beta\mathbb{E}[R]}$. Substituting $\sigma = \beta^{-(1+\eta)/4}$:

$$\beta\,\mathbb{E}[R] \sim \beta\sigma^4 d^2 = \beta^{-\eta} d^2 \;\to\; 0 \qquad \Rightarrow\ \eta > 0 \text{ (remainder must vanish)}, \tag{11}$$

$$\beta\sigma^2 = \beta^{(1-\eta)/2} \;\to\; \infty \qquad \Rightarrow\ \eta < 1 \text{ (bias must survive)}. \tag{12}$$

If σ does not shrink as β grows, the remainder blows up exponentially; if it shrinks too fast, the regularization strength $\frac{\sigma^2}{2}\operatorname{tr} H$ is washed out by the large β. Each end of $\eta \in (0,1)$ blocks one failure mode—σ must move together with β, at a rate caught in between. That is the entire content of the coupling.

4.4 Strengths and Weaknesses

The advantage of fSGLD can be summarized as "two things for free": with a single gradient and no auxiliary variables, it obtains both global exploration (inherited from SGLD) and a flatness bias (the implicit $\operatorname{tr} H$ regularization), without adding

hyperparameters. I see three limitations. First, the regularization strength is welded to the temperature by the coupling ($\beta\sigma^2 = \beta^{(1-\eta)/2}$): a strong flatness bias requires a large β. Second, $\operatorname{tr} H$ is not invariant under reparametrization of the network [10], and the isotropic perturbation $\mathcal{N}(0, \sigma^2 I_d)$ treats parameter layers of widely different scales uniformly. Third, I have not yet studied the convergence proofs and the experiments in detail, which I leave for subsequent reading.

5 Overall Comparison

Seen side by side, the four algorithms trace an evolution in the attitude toward noise. GD has no noise: exact, expensive, and purely local, it can only reach a stationary point of the basin it starts in. SGD tolerates noise: each step costs a single gradient, but the iterate converges merely to a point within an $O(\rho^2)$ neighborhood unless the stepsize is diminished. SGLD designs noise: temperature buys provable global exploration, and the output is no longer a single point but a distribution $\pi \propto e^{-\beta u}$, yielding uncertainty and out-of-distribution detection as a byproduct. fSGLD endows noise with geometric structure: a perturbed gradient redirects the target to $e^{-\beta v}$, buying flat solutions at essentially unchanged cost and with not a single additional hyperparameter. Each upgrade is a small edit—SGD changes how much data enters the gradient, SGLD adds one Gaussian term, fSGLD moves one evaluation point—yet each yields something qualitatively new. For me, this thread is the main takeaway from the two documents.

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